Comprehensive Assessment of Scope 1 and Scope 2 Carbon Emissions and Sustainability Roadmap for Union Bank PLC

Project Proposal

FS 3002 - Service Learning

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01.Abstract

Climate change has become a critical global issue, generating environmental, economic, and social challenges worldwide. According to the Intergovernmental Panel on Climate Change, rising greenhouse gas (GHG) concentrations are primarily driven by human activities. In response, organizations are increasingly expected to monitor, report, and reduce their carbon emissions. Integrating sustainability into corporate strategies has become essential for aligning with global climate goals and ensuring long-term resilience.

A carbon footprint represents the total greenhouse gas emissions produced directly or indirectly by an organization, expressed in carbon dioxide equivalents (CO_2e). The internationally recognized Greenhouse Gas Protocol classifies emissions into Scope 1 (direct emissions), Scope 2 (indirect emissions from purchased electricity), and Scope 3 (other indirect value-chain emissions). In service-sector institutions such as banks, Scope 1 emissions mainly arise from generator fuel use and refrigerant leakage, while Scope 2 emissions largely result from electricity consumption in offices and branches.

Although banks are not heavy industrial polluters, their operational activities and financial influence significantly impact climate-related outcomes. Sustainability in the banking sector involves reducing operational emissions, promoting green financing, and integrating Environmental, Social, and Governance (ESG) principles into decision-making. Conducting carbon footprint assessments improves operational efficiency, strengthens corporate reputation, and aligns institutions with international sustainability standards.

Accurate carbon accounting depends on reliable activity data and appropriate emission factors. Guidelines provided by the Intergovernmental Panel on Climate Change support standardized emission estimation based on fuel and electricity consumption. However, localized emission factors are crucial, particularly in countries such as Sri Lanka, where the electricity generation mix—including hydro, thermal, and renewable sources—affects carbon intensity. Applying context-specific factors enhances the accuracy and policy relevance of corporate carbon assessments.

Statistical analysis further strengthens environmental performance evaluation. Techniques such as One-Way ANOVA help determine whether significant differences exist between monthly emission levels, enabling organizations to identify high-emission periods and optimize energy use. Analytical tools such as Minitab and SPSS support evidence-based decision-making, contributing to more effective and targeted sustainability strategies.

02. Introduction

In the current international situation of climate change and environmental destruction, the organizations are being more and more held accountable to observe the quantity of emission of carbon. The energy consumption and use of resources by businesses makes this one of the important steps toward the attainment of environmental sustainability. Greenhouse Gas (GHG) emissions measurement also allows organizations to know how their activities affect the environment, and what can be done to enhance energy efficiency, decrease emissions and pursue more sustainable practices. With the rising global sustainability systems and the national environmental commitment, corporate organizations are currently emphasizing the low-carbon approach and accountability in the management of resources as their long-term objectives of operation.

The project is aimed at determining the carbon footprint of the operational practices of the Union Bank PLC that includes its Head Office and 61 branches that are established in Sri Lanka. The research is designed to effectively gather and evaluate information about key sources of emissions including electricity consumption, generator fuel usage, water usage and other operation activities. These sources are sources of Scope 1 and Scope 2 emissions, which constitute major part of the total carbon footprint of the bank. The project will demonstrate a general picture of the current environmental impact of the bank by measuring the levels of emissions through the right way of doing it as well as standardized methods.

One of the purposes of this evaluation is to determine the areas with high consumption, odd patterns of use, and inefficient operations that result in higher emissions. The study of the trend of energy and resource consumption involved in various regions and over time will assist in making evidence-based decisions and being able to focus on areas of improvement. The results of this paper will help in coming up with effective solutions to the problem by facilitating the formulation of feasible solutions that will help save energy, streamline resource management, and have the organization adopt environmentally friendly operation practices.

Moreover, the long-term vision of the bank to stop relying on fossil fuels and be engaged in sustainable and renewable sources of energy has aligned this project. The research will facilitate long term planning in the reduction of carbon emissions through energy conservation, implementation of renewable energy technologies, and sustainable development of infrastructure. Finally, the industrial sustainability study is likely to enhance the achievement of corporate sustainability objectives, corporate environmental responsibility, and improve the organizational adherence to responsible and sustainable banking practices.

03.Research Objectives

3.1 General Objective

To assess and analyze the operational carbon footprint of **Union Bank PLC** by quantifying greenhouse gas emissions and identifying opportunities for emission reduction and sustainable resource management.

3.2 Specific Objectives

1. **Emission Quantification**

   To estimate the annual greenhouse gas emissions (CO_2e) associated with operational activities, with a primary focus on Scope 1 (direct emissions from fuel and generator usage) and Scope 2 (indirect emissions from electricity consumption) across the Head Office and all branch locations.)

2. **Emission Hotspot Analysis**

   To identify branches, operational areas, and time periods (monthly or seasonal) that contribute the highest levels of emissions in order to understand spatial and temporal variations in energy consumption and carbon output.

3. **Trend and Pattern Analysis**

   To analyze electricity, water, and fuel consumption patterns over time to detect abnormal increases, inefficiencies, and potential areas for energy optimization.

4. **Sustainability Strategy Development**

   To develop evidence-based recommendations and a practical sustainability roadmap aimed at reducing greenhouse gas emissions, improving energy efficiency, and optimizing operational costs.

5. **Baseline Establishment for Future Monitoring**

   To establish a reliable carbon emissions baseline that can be used by the organization for future tracking, reporting, and evaluation of emission reduction initiatives.

04. Methodology

This study adopts a quantitative and analytical research approach to assess greenhouse gas (GHG) emissions generated from the operational activities of **Union Bank PLC**. The methodology is designed to ensure systematic data collection, accurate emission estimation, and reliable statistical analysis to support evidence-based sustainability recommendations.

4.1 Data Collection and Preparation

Primary operational data for the year 2025 will be obtained from the Finance Section for the Head Office and all 61 branches. The dataset will include monthly electricity consumption (kWh), generator fuel usage (liters), and air-conditioning refrigerant or maintenance records (kg). These data represent the major sources contributing to operational carbon emissions. Prior to analysis, the data will be cleaned and validated by checking for missing values, inconsistencies, and outliers. Standardization procedures will be applied to ensure uniform units and consistency across all branch records.

4.2 Emission Estimation Framework

Greenhouse gas emissions will be calculated in accordance with internationally recognized standards such as the Greenhouse Gas Protocol and IPCC-based emission factors. Scope 1 emissions will be estimated from direct sources, including generator fuel consumption and refrigerant losses. Scope 2 emissions will be calculated from purchased electricity using the national grid emission factor. All activity data will be converted into carbon dioxide equivalent (CO_2e) to allow standardized comparison across branches and time periods. Monthly and annual emission totals will be aggregated to develop a comprehensive emissions inventory.

4.3 Statistical Analysis

Data analysis will be conducted using **IBM SPSS Statistics** and **Minitab**. Descriptive statistical techniques, including mean, standard deviation, and percentage contribution analysis, will be used to summarize emission levels and identify major emission sources. Time-based trend analysis will be performed to examine seasonal variations and detect unusual increases in emissions. A One-Way ANOVA will be applied to determine whether there are statistically significant differences in carbon emissions across months. Where significant differences are observed, post-hoc tests will be conducted to identify specific high-emission periods. In addition, branch-wise comparisons and correlation analysis will be used to identify high-emission locations and examine relationships between resource consumption variables.

05. Expected Outcomes

This research project will yield feasible and high-level results that will assist the sustainability objectives of Union Bank PLC besides enhancing the performances of operations and environmental responsibility.

5.1 Operational Efficiency

The analysis will establish the trends in the energy usage patterns at the monthly level and the level of the branches so that high-use periods, inefficiencies, and anomaly in the variations could be detected. This knowledge will assist the bank to adopt specific behaviors in energy management to cut down on avoidable use of electricity and fuel hence cutting down on waste in operations and reducing the cost of energy.

5.2 Corporate and Strategic Sustainability Alignment.

The results will enable the bank to improve its environmental management strategy and increase its corporate sustainability profile. Measurement of Scope 1 and Scope 2 emissions and important emission sources will help the study to guide the management in association of operational decisions made by the firm in the long run sustainability goals and enhance green corporate image of the organization among the stakeholders.

5.3 Climate risk preparedness and regulatory preparedness.

The project will help in setting up an organized carbon emissions base and reporting system that can be utilized to conduct further monitoring and subsequent sustainability reporting. This will help the bank to be proactive to changing environmental laws, climate disclosure demands, and sustainability guidelines in the country or internationally, thus enhancing its climate resilience in the long-term.

5.4 Emission Reduction Initiatives Decision Support.

The findings will offer an evidence-driven perspective in relation to new endeavors like the implementation of energy efficiency, optimization of the use of generators, and the gradual implementation of renewable energy solutions. By following these recommendations, the bank will be moving forward to the low-carbon operations and sustainable resource management.

06. Project Timeline

Phase 1 - Week 1: Consultation with the stakeholders and project planning activities were performed, which involved attending a climate change awareness program and discussions about the project scope, objectives, and expectations.

Phase 2 - Week 2: The data on climate-related information of Annual Reports of the chosen companies was examined and offered. An awareness session on climate was also conducted in the team to increase awareness of sustainability practices.

Phase 3 - Week 3: The Head Office and 61 branches were involved in primary data collection (Session 1), which involved gathering data related to the operations of the company including electricity consumption, fuel use and other pertinent data.

Phase 4 - Week 4: Data collection (Session 2) was monitored and concluded on the Head Office and branches and data verification and validation done to guarantee the accuracy and completeness.

Phase 5 - Week 5: Trend analysis and One-Way ANOVA were used in statistical analysis and estimation of monthly emittance and in IBM SPSS Statistics and Minitab to evaluate significant variations and periods of high emission rates.

Phase 6 - Week 6: It was a formal meeting where the issues of Small and Medium Enterprise (SME) and value chain emissions were discussed and how they would be considered in the assessment of the overall carbon footprint.

Phase 7 - Week 7: The results of the analysis were used to generate sustainability recommendations, which concentrated on the possibility to reduce the emission and improve operations.

Phase 8 - Week 8: The final report was to be made and presented and the project findings and recommendations presented to the supervisors and stakeholders.

07. References

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